

TECHNICAL SKILLS

Programming Languages/Frameworks/Libraries/Design Software

- Languages/Frameworks: Python, SQL, C++, Javascript, Typescript, Next.js, React.js, HTML, Tailwind CSS, FastAPI, LangChain, Node.js, Swift, Kotlin
- Cloud/Infrastructure: AWS (RDS, Elastic Beanstalk, ECR), Docker, Cloudflare, Celery, Redis
- Other: Git (GitLab/GitHub), Framer Motion, PostgreSQL, RESTful APIs, Gtest, Linux/Unix OS, TestRail, ADB, Golang

EXPERIENCE

Ford Motor Company

Waterloo, ON

Software Engineer - Product Development

January 2026 - April 2026

- Built Python automation to eliminate manual pre-setup for security test suites, automating Wi-Fi configuration, user profile creation, and Google Play sign-in via ADB UI inspection, reducing setup time to zero
- Designed and authored 8+ manual test cases for the VHAL Simulator callback feature, covering TX signal registration, real-time callback verification, and signal unregistration across 8 supported climate control signals
- Developed cross-platform automated sanity tests and TestRail documentation for Ford's Key Management Trusted Application (Android & QNX), validating secure key encryption, decryption, and lifecycle management end-to-end
- Built two reusable helper functions for asserting Suspend-to-RAM state transitions using SSH timeout behaviour as a proxy for device sleep, enabling consistent, reliable verification across test benches
- Executed and validated automated test suites for the VHAL Simulator tool across multiple release versions using the Slash framework

TranQuility Inc.

Toronto, ON

Software Engineer — Full Stack Development

May 2025 - August 2025

- Led 0→1 development of Encubatorr, an AI-powered consultation platform, owning the full-stack architecture from an empty repo to a production-ready staging environment
- Designed and deployed a staging environment on AWS using Docker, Elastic Beanstalk, and RDS, enabling the team to confidently move toward a first production release
- Integrated LangChain conversation memory to give the AI consultation engine cross-session context, eliminating repetitive inputs and directly improving early user satisfaction
- Implemented asynchronous task processing with Celery + Redis, improving platform responsiveness and reliability under concurrent load
- Designed a secure auth system with bcrypt hashing, session management, Google OAuth, and email verification, handling security end-to-end without third-party auth providers
- Built a responsive React.js dashboard with real-time AI chat and multi-step onboarding forms, reducing friction in early prototype evaluations
- Wrote SQL in PostgreSQL to design and set up the Encubatorr database, including schema creation and a data dictionary for data definitions

Projects

Automated Python Email Sender

- Built an automated email system for non-profit Coding Pals using smtplib and CSV-driven templates, reducing manual email handling by 80% and enabling personalized communication to 500+ recipients per campaign
- Reengaged 300+ past participants post-pandemic, contributing to a 20% increase in class registrations and measurably improving community outreach efficiency

UW Robotics Design Team

Waterloo, Ontario

Software Team

September 2024 - April 2025

- Built modular ROS2 applications, working hands-on with node communication, real-time data handling, and inter-process messaging for autonomous robotics systems

EDUCATION

University of Waterloo

Software Engineering, Bachelor of Software Engineering

September 2024 - April 2029